

Evaluation Report

For the performance metrics, Buyer and Seller clients were simulated by spawning them on different threads. For every run, the Buyer and the Seller thread first run the register account operation and then the login operation. After that, for the rest of the runs, the buyer client iterates through the following actions: "PRODUCT_SEARCH", "GET_BUYER_PURCHASES", "SAVE_CART", "CLEAR_CART", "DISPLAY_CART", and the seller client iterates through the following actions: "GET_RATING", "REGISTER_PRODUCT", "CHANGE_ITEM_PRICE", and "GET_RATING". At the end of each run, the clients perform the logout operation. This is similar to the previous experiments with TCP, the only change is we are making rest calls

The operations were performed on the google cloud instances: total 6 instances, each for buyer client, seller client, their middlewares and the gRPC database services. Each VM is an e2-medium machine (2vCPUs, 4GB RAM).

The database was deployed using the cloud SQL instances.

Scenario	Average Response Time (ms)	Throughput
1 Concurrent Buyer and Seller	10.42	178.01 req/s
10 Concurrent Buyer and Seller	55.27	358.83 req/s
100 Concurrent Buyer and Seller	659.4	286.95 req/s

As we move from 1 concurrent Buyer and Seller to 10, we can see that the average response time increases as the number of concurrent requests increases, and that also leads to increased throughput from scenario 1 to 2. When we move to scenario 3, the average response time is much higher, and we witness a drop in throughput. This indicates that handling 200 total concurrent users in the current situation is difficult.